All solutions to graded problems must be typed in LaTeX. All other problems can be typed or written, but must be completed. Graded problems must be organized in numerical order. All problem sets (both graded and non-graded) must be uploaded to Canvas as a single PDF document. Students can find the grading scale in our Syllabus.

Graded

Exercise 1. Let X and Y be topological spaces.

1. Let $p : X \to Y$ be a continuous function. Show that if there is a continuous function $f : Y \to X$ such that $p \circ f$ equals the identity function on Y, then p is a quotient map.

2. Provide an example of X, Y, p showing that continuity of f is necessary.

Exercise 2. For a topological space X, the *cone over* X denoted CX is the quotient space of $X \times [0, 1]$ corresponding to the partition consisting of all one-point sets in $X \times [0, 1)$ and the set $X \times \{1\}$.

Similarly the *suspension over* X denoted ΣX is the quotient space of $X \times [0, 1]$ corresponding to the partition consisting of all one-point sets in $X \times (0, 1)$ and the sets $X \times \{0\}$ and $X \times \{1\}$.

Let $S^1 = \{x \times y \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, $\mathbb{D} = \{x \times y \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, and $S^2 = \{x \times y \times z \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ be considered with their respective subspace topologies from $\mathbb{R}^n$.

1. Prove that CS^1 is homeomorphic to $\mathbb{D}$.

2. Prove that ΣS^1 is homeomorphic to S^2.

3. Use your answers to 1 and 2 to prove the product space $S^1 \times (0, 1)$ (the open annulus) is homeomorphic to both $\text{Int}(\mathbb{D}) - \{0 \times 0\}$ and $S^2 - \{0 \times 0 \times 1, 0 \times 0 \times -1\}$.

Non-Graded

Exercise 3. Let $\pi_1 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the projection on the first coordinate. Define

$$A = \{x \times y \in \mathbb{R}^2 : x \geq 0 \vee y = 0\}.$$

Let $q = \pi_1|_A$ be the restriction of π_1 to A. Show that q is a quotient map that is neither open or closed.

Exercise 4. Define $\mathscr{L} \subset \mathbb{R}^2$ to be the subset

$$\mathscr{L} = \{x \times y \in \mathbb{R} : y^2 - x^2 + x^4 = 0\}.$$

Let $X = [0, 2]$ and suppose X^* is the partition of X consisting of one-point sets in $X - \{0, 1, 2\}$ and the set $P = \{0, 1, 2\}$. Show that $\mathscr{L}$ with the subspace topology in $\mathbb{R}^2$ is homeomorphic to X^* with the induced quotient topology.